

Chemical Engineering

Comprehensive Guide

Professional Edition 2026

Harz Digital Services

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Chapter 1

Introduction to Chemical Engineering

Chemical engineering combines chemistry, physics, mathematics, and economics to design and operate processes that convert raw materials into useful products. It is the engineering of transformation -- turning raw materials into fuels, medicines, materials, food, and chemicals.

Core Principles

Chemical
processe

Key Areas

Chemical

Industries Served

Oil and g
consume

The Chemical Engineer's Toolkit

Mass an
software

Why Chemical Engineering Matters

From the
chemical

Chapter 2

Material and Energy Balances

Material and energy balances are the foundation of chemical engineering. They are the accounting of all mass and energy entering, leaving, and accumulating in a process.

Law of Conservation of Mass

Mass can

A reactor feeds 100 kg/h of reactant A. If 80% converts to product and 20% leaves unreacted, the output is 80 kg/h product + 20 kg/h unreacted A = 100 kg/h total.

Material Balance Procedure

1. Draw a

Types of Balances

Overall b

Energy Balance (First Law of Thermodynamics)

Energy in

A heat exchanger cools 1000 kg/h of water from 80C to 30C. Heat removed = $m \times c_p \times \Delta T = 1000 \times 4.186 \times (80-30) = 209,300 \text{ kJ/h} = 58.1 \text{ kW}$.

Recycle and Bypass

Recycle s

Purge

Purge str
gaseous

Chapter 3

Chemical Reaction Engineering

Chemical reaction engineering is the study of chemical reactions and reactor design. It combines reaction kinetics with reactor engineering to determine optimal conditions for industrial reactions.

Reaction Rate

Rate of r

Reactor Types

Batch Re
chemical

CSTR Design Equation

$V = F_{A0} / r$

PFR Design Equation

$V = F_{A0} \int_0^X \frac{dX}{r}$

A first-order liquid reaction ($k=0.1/s$) in a CSTR with feed $0.01 \text{ m}^3/s$, $C_{A0}=2 \text{ mol/L}$, desired conversion= 0.8 . $V = F_{A0} \times X / (k \times C_{A0} \times (1-X)) = 0.01 \times 0.8 / (0.1 \times 2 \times 0.2) = 0.2 \text{ m}^3$.

Catalysis

Catalysts
(phase), h

Non-Ideal Reactors

Real reac
Models: c

Chapter 4

Separation Processes

Separation processes purify products and isolate desired components from mixtures. They often represent the majority of capital and operating costs in chemical plants.

Distillation

The most

A distillation column separates 50/50 benzene/toluene at 1 atm. Benzene is more volatile (bp 80C) and goes to the top. Toluene (bp 111C) exits the bottom.

Absorption and Stripping

Absorption

Extraction

Liquid-liq

Membrane Separations

Reverse

Crystallization

Forms pu

Adsorption

Surface-l
separation

Chapter 5

Transport Phenomena

Transport phenomena describes the transfer of momentum, energy, and mass in chemical processes. These three transport processes are governed by analogous equations.

The Three Transports

Momentu

Dimensionless Numbers

Reynolds

Convective Transport

Heat: Nu

Water (Pr=7, Re=10,000) in a pipe. $Nu = 0.023 \times (10000)^{0.8} \times (7)^{0.4} = 0.023 \times 1585 \times 2.18 = 79.3$. $h = Nu \times k/D = 79.3 \times 0.6/0.05 = 952 \text{ W/m}^2\text{K}$.

Interphase Mass Transfer

Two-film

Applications

Pipeline
design.

Chapter 6

Process Control and Instrumentation

Process control maintains process variables (temperature, pressure, flow, level, composition) at desired values despite disturbances. It is essential for safe, efficient, and profitable plant operation.

Control Loop Components

Sensor: M

Feedback Control

The most

Feedforward Control

Measure
accurate

Cascade Control

Uses nes
controller

Advanced Control Strategies

Ratio con

Process Safety Instrumentation

Safety In
Level) de

A reactor temperature exceeds the high-high limit. SIS triggers emergency cooling and isolation valves, independent of the basic process control system.

Chapter 7

Process Safety and Hazard Analysis

Process safety prevents catastrophic incidents in chemical plants. The lessons from Bhopal, Piper Alpha, and Texas City have shaped modern safety practices.

Hazard Identification

HAZOP
reverse, c

Consequence Analysis

Dispersion
(Britter-M

Risk Assessment

Risk = Fr

Layer of Protection Analysis (LOPA)

Independ
Failure o

Inherently Safer Design

Minimize

Key Standards

OSHA P3

Chapter 8

Petroleum Refining and Petrochemicals

Petroleum refining converts crude oil into valuable products. It is one of the largest and most complex chemical engineering industries.

Crude Oil Composition

Crude oil
measured

Refining Processes

Distillation
gas oil, fuel

A refinery processes 100,000 barrels/day of crude. Typical yield: 15% LPG, 35% gasoline, 20% diesel, 10% jet fuel, 20% fuel oil and others.

Petrochemicals

Ethylene
polyethylene

Downstream Products

Polymers

Chapter 9

Green Engineering and Sustainability

Green engineering applies environmental principles to chemical engineering design. It aims to minimize environmental impact while maintaining economic viability.

Principles of Green Engineering

1. Design

Circular Economy

Move from linear processes

Carbon Capture and Storage (CCS)

Post-combustion

Renewable Feedstocks

Bio-based

Water Treatment

Advanced

Life Cycle Assessment (LCA)

LCA evaluates carbon footprint

A bioplastic made from corn has lower carbon footprint than petroleum-based plastic, but may require more water and land. LCA helps compare total impact.

Future of Chemical Engineering

Green chemical processes